



Innovative Activity over the Business Cycle

Author(s): P. A. Geroski and C. F. Walters

Reviewed work(s):

Source: *The Economic Journal*, Vol. 105, No. 431 (Jul., 1995), pp. 916-928

Published by: [Wiley-Blackwell](#) for the [Royal Economic Society](#)

Stable URL: <http://www.jstor.org/stable/2235158>

Accessed: 05/09/2012 12:00

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at
<http://www.jstor.org/page/info/about/policies/terms.jsp>

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.



Wiley-Blackwell and Royal Economic Society are collaborating with JSTOR to digitize, preserve and extend access to *The Economic Journal*.

<http://www.jstor.org>

INNOVATIVE ACTIVITY OVER THE BUSINESS CYCLE*

P. A. Geroski and C. F. Walters

This paper examines cyclical patterns of innovative activity in the United Kingdom over the period 1948–83. Innovative activity turns out to have many of the properties of a random walk, but does show a tendency to cluster during booms. There is clear evidence of a long-term secular relation between the level of innovative activity and the level of economic activity, although this may have changed in the 1980s. Finally, the data provide no reason for thinking that these clusters of innovation cause cyclic variations in economic activity, but variations in economic activity do Granger cause changes in innovative activity.

Innovation is an integral part of the process of economic growth, but the mechanism which links innovation to the level of economic activity is not well understood. Some economists favour a ‘supply push’ model of innovation in which innovative activity is driven by exogenous and largely unpredictable advances in basic scientific knowledge. Supply push advocates often argue that innovative activity occurs in clusters, and many of them believe that fluctuations in innovative activity are an important cause of fluctuations in economic activity (a view which has also featured in the more recent real business cycles literature). Other economists subscribe to a ‘demand pull’ model of innovation in which changes in sales or profitability stimulate investments in R & D. In demand pull arguments, the important causal links usually (but not always) run from fluctuations in economic activity to innovation. Choosing between ‘supply push’ and ‘demand pull’ theories of innovation is, therefore, interesting for the light it sheds both on the determinants of innovation as well as on the determinants of cyclical fluctuations in demand.

In designing an experiment to discriminate empirically between these two views, we have been guided by three principles: the null hypothesis must have enough economic content to make its rejection interesting, the experiment must include large enough fluctuations in demand to enable one to spot what might be rather small demand pull effects and it must be reasonably robust to different definitions of ‘innovation’. The first principle is easy to satisfy, since supply push arguments often lead to a simple stochastic description of innovative activity. The second two have led us to track two measures of innovative activity (the number of major innovations produced in the United Kingdom, and the number of patents taken out by UK firms in the United

* We are obliged to the Gatsby Foundation and the Centre for Business Strategy for financial support; to Pari Patel, Keith Pavitt, John van Reenan and Geoff Dicks for help in assembling the data; to Anthony Garratt and Stephen Hall for technical assistance; and to Saul Estrin, Jonathan Haskel, John van Reenan, Dennis Mueller, Steve Klepper, Keith Pavitt, two referees, an Associate Editor, the Editor and seminar participants at the National Institute, the Institute for Fiscal Studies, Humboldt University, the Universities of Auckland, Waikato, Victoria, Nottingham, Birmingham, Exeter, the ANU, WIFO and SPRU for helpful comments on an earlier draft. The usual disclaimer applies, however.

States) through the cyclic upturns and downturns in UK economic activity over the period 1948–83. Our results suggest that major innovations and patents are pro-cyclical, and that the important causal relations run from variations in demand to variations in innovative activity but not from innovations or patents to changes in demand.

The plan of exposition is as follows. In Section I below, we review a range of hypotheses which suggest that innovative activity may be sensitive to business cycle influences. Section II describes our exploration of the data on major innovations and patents, while Section III summarises the results and links them to the various hypotheses discussed in Section I.

I. INVENTION AND IMPLEMENTATION CYCLES

‘Supply push’ summarises a range of determinants of innovative activity, including the evolution of scientific knowledge, the productivity of research labs, the opportunity costs of investing in new product or process development, and so on (see Rosenberg, 1974; Dosi, 1988; and others). One rather intuitively appealing (and commonly held) supply push view is that increases in the stock of knowledge occur randomly over time, causing the production of innovations to vary exogenously and unpredictably over time. At its simplest, this argument suggests that the number of innovations produced each year will follow a random walk over time. There is, however, at least one alternative characterisation of supply push that is worth considering. This argument turns on a distinction between major and minor innovations. Major innovations open up new technological trajectories, paving the way for a series of more minor innovations which explore the newly opened technological opportunities in waves of imitative entry. Jovanovic and Rob (1990) describe this process in terms of ‘intensive’ and ‘extensive’ search, but one can also think about it as reflecting economies of scale in the relationship between inputs of knowledge and the output of the innovation process. When economies exist, major advances in scientific knowledge will have more than proportionally larger effects on the output of innovations than smaller advances in knowledge, not least because each major innovation triggers a wave of subsequent minor innovations. As with many other forms of economic activity that are subject to increasing returns, this argument predicts that variations in inputs (i.e. in the stock of knowledge) will lead to a more variable stream of output (i.e. innovations) over time; i.e. that innovations will appear in clusters.¹

‘Demand pull’ influences on innovative activity include a range of effects driven by changes in consumer behaviour, the competitive structure of markets

¹ To make these arguments more concrete, consider the following simple illustrative model. Let n_t be the log of the number of innovations and s_t be the log of the stock of knowledge. If the two are related by a ‘production function’: $n_t = A + \alpha s_t$ and if the stock of knowledge varies unpredictably over time according to $s_t = s_0 + \epsilon_t + \theta \epsilon_{t-1} + \theta^2 \epsilon_{t-2} + \dots$, where ϵ_t is an iid random variable, then $\Delta n_t = \alpha \epsilon_t + \alpha(\theta - 1) \Delta n_{t-1} + \alpha \theta(\theta - 1) \Delta n_{t-2} + \dots$. If knowledge is long lived ($\theta = 1$) and constant returns prevail ($\alpha = 1$), then n_t follows a random walk. If $\theta = 1$ but $\alpha > 1$, n_t still follows a random walk but with a larger variance than if $\alpha = 1$. However, if $0 < \theta < 1$, then innovative activity displays an autoregression fluctuation overlaid on a random walk which will give the impression of a clustering of innovation at certain times.

and other factors which affect the valuation of inventions and/or the ability of inventors to appropriate the benefits created by them. Although the assertion that demand matters is controversial to some, what makes understanding the relationship between demand and economic activity really difficult is the diversity of views which exist about whether demand makes innovative activity pro or anti-cyclical, and about whether changes in demand are exogenous to the process of producing and using innovations. In what follows, we shall outline three arguments which suggest that inventions will appear in clusters timed with movements in demand. The first suggests that innovative activity may be counter cyclical, while the second and third both suggest that innovations are more likely to be produced in booms than in recessions.

The first argument turns on the opportunity costs of innovation. As is well known, established firms are likely to be slower to implement an invention than entrants if it threatens to displace their existing rents. The value of existing rents usually falls in a recession, and if this decline is large relative to the returns to be gained from implementing a new product or process, firms will implement new inventions during cyclical downturns.² Further, any time a firm changes or expands its set of activities, it must divert resources from current activities (e.g. production) to effect the necessary changes in its strategy and structure (for a classic discussion, see Penrose (1959)). An attractive time to do this is during a recession when current activities are relatively less profitable, and this is especially the case for reorganisations which do not require large amounts of cash (see Hall (1991) and others who have argued that firms have strong incentives to make investments in 'organisational capital' during recessions). For one or both of these reasons, then, one might expect to observe counter-cyclical variations in innovative activity (see also Mensch, 1979; Kleinknecht, 1987; and others).

The second demand pull argument focuses on the limited ability of markets to absorb new products at any one time (see Judd (1985) for a formal model which explores this argument). Each new product introduced at time t competes with a range of existing products, and each is likely to be followed by a wave of imitative or 'me-too' products. As more and more new products are introduced in period t , competition for the same consumer spending reduces the profitability of each, and, as a consequence, the flow of new products brought to the market will dry up. The implementation of new inventions is likely to resume when economic growth generates enough market expansion to absorb more new products, and, indeed, the rush to take advantage of this limited window of opportunity is likely to lead to a clustering of new innovations introduced into the market to take advantage of the expansion of demand. Note that this argument, like the first, posits a causal link from changes in demand to variations in implementation activity over time.

² Arrow (1963) is a classic statement of the rent displacement argument. In this context, it is not entirely persuasive for two reasons. First, it is not entirely obvious why the relative value of existing rents should necessarily fall in a recession, and, second, if new innovations complement existing activities (rather than displacing them), then established firms are likely to innovate before entrants (see Gilbert and Newbery, 1982).

The third demand pull argument starts with the observation that appropriability problems mean that inventors will have only a limited window of opportunity in which to generate profits from their innovative activities. As a consequence and regardless of when they produce an innovation, firms will wish to implement their innovations when market conditions are most favourable. This is, of course, likely to be during periods of high and growing demand. Note that the causal links between innovation and demand in this argument are rather subtle. If what drives the implementation of inventions is expectations about near future booms in demand and if innovators act on these expectations, then their activities may become self-fulfilling (Schleifer, 1986). That is, clusters of innovative activity may generate booms in demand.

Thus, to explore the role of demand on innovation, we adopt the null hypothesis that innovative activity is supply driven. In practice, this means that we expect to observe the number of innovations produced to follow a random walk over time. Departures from a random walk are not necessarily inconsistent with supply push theories of innovation, since increasing returns and other supply side effects may result in a clustering of innovations produced over time. However, if these clusters are systematically linked with changes in demand and if, in addition, demand changes appear to cause changes in the level of innovative activity, then it would be difficult to avoid concluding that demand mattered. How it matters is the second question of interest. Although the various demand pull arguments which we have summarised are reasonably persuasive on the proposition that demand matters, they do not add up to a strong *a priori* case for believing that innovative activity will necessarily be pro-cyclical.

II. POST-WAR INNOVATIVE ACTIVITY IN THE UNITED KINGDOM

II.1. *The Data*

The first set of data that we shall use is a count of the major innovations produced and used in the United Kingdom over the period 1948–83. The data were generated by the Science Policy Research Unit (SPRU) at the University of Sussex as part of a large-scale study into the sources and incidence of innovation in the United Kingdom during the post-war period, and cover 4,258 innovations in total (see Townsend *et al.*, 1981). To qualify for inclusion in the data, an innovation had to be judged to be both a commercial success and a technological breakthrough. Fig. 1 displays the total number of innovations over time together with an index of industrial production, suitably scaled. It shows that the ebb and flow of innovative activity is highly irregular over time, with three and four-year clusters of innovative activity overlaid on an upward trend that comes to an abrupt halt at the end of the 1970s, near the end of the period covered by the data. The average number of innovations produced each year over the period is 118, ranging from a minimum of 39 in 1949 to a maximum of 195 in 1970. It seems to be the case that the number of innovations produced each year varies pro-cyclically, not least because both

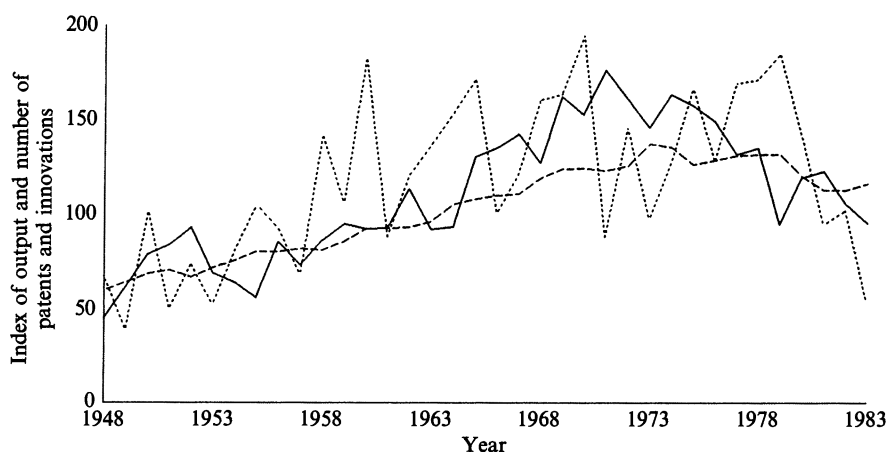


Fig. 1. Innovations and patents in the United Kingdom, 1948–83. —, patents; ---, index of manufacturing output;, innovations.

series display an upward trend until the late 1970s and both turn down sharply at the beginning of the Thatcher years. The partial correlation between the two is 0.6229, but changes in innovative activity are much more weakly correlated to changes in industrial production (the correlation is 0.2298).

The second data set that we shall use consists of the number of patents taken out in the United States by UK firms over the same period (also kindly provided by SPRU). Fig. 1 also shows the total number of patents per year over time suitably scaled. The level of patenting activity by UK firms generally rose over the period, and also appears to vary pro-cyclically. The average number of patents per year is 2,225, varying from a minimum of 896 in 1948 to a maximum of 3,533 in 1971. The partial correlation between patents and industrial production is 0.8624, while that between their first differences is -0.1763 . The correlation between patents and innovations is 0.5792 (in first differences, it is 0.0192).

The data have two flaws in their construction which may have some effect on the outcome of our experiments. First, there are grounds for thinking that part of the fall-off in the number of major innovations produced after 1980 is due to a winding-up of the data collection process undertaken by SPRU. Further, post-1979 innovations will have had a much smaller time in which to establish their commercial worth and, as a consequence, fewer of them are likely to be recorded in the data (*ceteris paribus*). Second, manpower shortages in the US patents office induced an artificial drop in patenting in 1979 which lasted for a year or two before patenting returned to 'normal' levels.³ Both of these problems impart a potentially pro-cyclical bias to our tests, and to explore

³ Indeed, Griliches (1990) has shown that short-run fluctuations in patents granted are more closely related to employment in the patent office than to patent applications.

the effect that this might have on our results, we will repeat our calculations for the truncated sample 1948–79.⁴

II.2. *The Experiment*

The basic structure of the experiments which we will apply to this data set is straightforward. Since the null is that innovative activity follows a random walk over time, we will start by applying unit root tests to both the patents and the innovations data. These tests are interesting both as characterisation of supply push models of innovation, and as a guide to whether we should model innovations and patents in levels or first differences. The next step is to ascertain whether supply side factors cause a clustering of innovations or patents over time, and we will do this by fitting simple autoregressive models to each series. We then introduce demand into the experiment, first testing whether changes in demand Granger cause variations in innovations or patents or *vice versa*, and then examining whether demand and innovations or patents are co-integrated. Both of these tests will help us determine the appropriate way to generalise the autoregressive representation of innovations and patents to include demand.

II.3. *The Results*

Our null hypothesis is that the number of inventions produced and implemented follows a random walk over time, and it appears to be broadly consistent with the data. Inspection of the autocorrelations of n_t and p_t (the log of the number of innovations and patents respectively) and Δn_t and Δp_t (the first difference of the logs) for up to 20 periods reveals that both innovations and patents are highly serially correlated.⁵ Further, unit root tests applied to both sets of data fail to reject the null of a unit root in n_t and in p_t . These conclusions are each more marked when the post-1979 observations are dropped. Both first differenced series are, however, stationary, although both show enough systematic movement over time to lead us to reject the null hypothesis that (the log of) innovations and patents follows a random walk.⁶

⁴ Another potential problem arises from the fact that the SPRU innovations data which we are using provide a record only of successful innovations. If the major determinant of success is that an innovation is introduced in a boom, then the way that the SPRU data have been assembled (i.e. the focus on successful innovations) may lead to a sample selection bias against the 'supply push' hypothesis. This is unlikely to be true for patents.

⁵ To test the null hypothesis that all the autocorrelation coefficients for the innovation series are jointly zero, we used the Ljung–Box modified Q statistic. For the log level of innovation (lag order in parentheses) LB = 7.08 (1), 25.51 (4), 31.72 (8), 37.88 (12) and 54.68 (16), all of which are more than twice their 10% χ^2 critical values. In first differences, LB = 13.7 (1), 20.8 (4), 25 (8), 57.3 (12), 62.2 (16) and 66.9 (20) respectively, which also exceed their critical values. For the log of patents, LB = 23.5 (1), 69.7 (4), 103 (8), 106 (1), 121 (16) and 168 (20), all of which are several times higher than their critical values. In first differences, LB = 1.45 (1), 2.18 (4), 6.84 (8), 12.9 (12), 20.8 (16) and 23.4 (20), all comfortably below their critical values.

⁶ We computed (augmented) Dickey–Fuller tests with up to 12 lags and a constant, the residual based test suggested by Phillips (1987) and Phillips and Ouliaris (1990) and Cochrane's (1988) non-parametric statistic, and obtained similar results across methods (for surveys of unit root tests, see Campbell and Perron (1991) and Banerjee *et al.* (1993)). For innovations (p values in parentheses), $DF(34, 1) = -1.782$ (0.715), while in

We ran a range of autoregressions in Δn_t and Δp_t up to order five for both variables, but found no evidence to suggest that higher-order models are superior to an AR(1) model. In fact, AR(1) models provide a fairly accurate and parsimonious representation of the data, but the fit of the AR(1) and AR(2) models for both innovations and patents is poor ($R^2 < 0.31$).⁷ As a consequence, it is difficult to make an overpoweringly strong case for rejecting the null of a random walk, at least over a short to medium run time horizon.

Nevertheless, there is evidence in the data of some clustering of innovative and patenting activity above and beyond that created by a random walk, and it is clearly possible that this clustering is linked to variations in demand. Measuring the level of demand, y_t , by the log of the index of production, and denoting its first difference as Δy_t , we found good reasons for thinking that y_t is not stationary, but that Δy_t is.⁸ We then examined the causal relations between Δn_t , Δp_t and Δy_t , and found fairly strong evidence to support the assertion that output Granger causes innovations, but no evidence whatsoever to suggest that innovations Granger cause variations in output. Similarly, output appears to Granger cause patents, but patents do not Granger cause output. Further, there is no evidence of a causal relation running from innovations to patents, although there is some rather weak evidence that lagged ($t-2, t-3$) patents positively affect innovations.⁹ Finally, since output depends on the stock of knowledge which is, in turn, affected by innovative activity, we tested for the existence of co-integrating relations between output and innovations/patents. The evidence on the relationship between innovations and output is a little mixed, but suggests that innovations and output are probably co-integrated. It is clearly the case that patents and output are co-integrated, while patents and innovations are not.¹⁰ These test results, coupled with the clear and very robust positive correlations which exist between y_t and

first differences, $DF(33, 1) = -4.29$ (0.009). For patents (p values in parentheses), $DF(36, 1) = -2.771$ (0.223), Phillips $Z(t) = -2.183$ (0.224) and $Z(t \text{ hat}) = -2.567$ (0.311). In first differences $DF(35, 1) = -6.8032$ (0.000). For the 1948–79 sample of patents; $DF(32, 1) = -1.943$ (0.633), $Z(t) = -2.095$ (0.256) and $Z(t \text{ hat}) = -2.412$ (0.385), while in first differences $DF(31, 1) = -6.349$ (0.000).

⁷ The AR(1) model of innovations is: $\Delta n_t = -0.5493\Delta n_{t-1}$, with a t-statistic on the lagged dependent variable of -3.726 , $R^2 = 0.309$ and $LLH = -8.428$. The AR(2) model is: $\Delta n_t = -0.5423\Delta n_{t-1} + 0.0102\Delta n_{t-2}$, with t-statistics of -2.819 and 0.058 , $R^2 = 0.286$ and $LLH = -8.426$. An F test of the restrictions that $AR(2) = AR(3) = AR(4) = AR(5) = 0$ is accepted with $F = 1.366$ (0.272). For patents, the AR(1) model is: $\Delta p_t = -0.645\Delta p_{t-1}$, with a t-statistic of -0.627 , $R^2 = 0.006$ and $LLH = 46.256$. The AR(2) model is: $\Delta p_t = -0.0719\Delta p_{t-1} - 0.1117\Delta p_{t-2}$, with t-statistics of -0.699 and -1.087 , $R^2 = 0.011$ and $LLH = 46.857$. Restrictions analogous to the innovations autoregressions are also accepted with $F = 1.384$ (0.246).

⁸ For earlier work on this subject, see Campbell and Mankiw (1987), Cochrane (1988) and the overview in Stock and Watson (1988); our results are similar to those reported by Mills (1991) and Lee *et al.* (1992) for the United Kingdom.

⁹ In all cases, we estimated a range of models, including as many as 5 lags in output and innovations/patents, with and without constants and time trends, and invariably came to the same conclusion about Granger causality. The results were qualitatively the same when we dropped the post-1979 data. Needless to say, these are bivariate causality tests and multivariate tests could yield different results.

¹⁰ We applied both Engle-Granger and Johansen tests to the data, estimating equations with and without constants and time trends, including lags and using output as both the 'dependent' and 'independent' variable. For innovations, we did not reject co-integration when output was the 'independent' variable, but did when it was the 'dependent' variable; co-integration tests on output and patents, and on patents and innovation were robust to this change. When we dropped the post-1979 data, we rejected co-integration between output and innovations (although the evidence against was not overpowering) and between patents and innovations, but we (tentatively) accepted co-integration for the equations linking output and patents.

n_t and y_t and p_t , lead us (with some slight reluctance) to accept co-integrating equations of the form:

$$n_t = -4.05 - 0.0264t + 2.108y_t, \quad (1)$$

(2.03) (2.214) (4.197)

where the absolute values of t statistics are shown in brackets beneath the estimated coefficients and $R^2 = 0.507$, and:

$$y_t = 1.741 + 0.015t + 0.311p_t, \quad (2)$$

(4.75) (8.862) (6.134)

where $R^2 = 0.924$.

All of these calculations suggest that an appropriate model of Δp_t should include lagged dependent variables, lagged values of Δy_t and an error correction term, while the analogous model of Δn_t might also include lagged values of Δp_t . In building these models, we followed the usual general-to-specific procedure, commencing from a specification that included up to five lags in each exogenous variable and then simplifying as appropriate. The result is

$$\Delta n_t = -0.009 - 0.3026\Delta n_{t-1} + 1.20\Delta y_{t-1} - 0.0620\Delta p_{t-1} - 0.783z_{t-1}, \quad (3)$$

(0.155) (1.66) (0.224) (0.215) (2.81)

with $R^2 = 0.589$, and

$$\Delta p_t = -0.0014 + 0.033\Delta p_{t-1} + 0.961\Delta y_{t-1} + 1.132z'_{t-1}, \quad (4)$$

(0.046) (0.211) (1.433) (2.405)

with $R^2 = 0.353$, where z_{t-1} in (3) is the residual from (1) and z'_{t-1} in (4) is the residual from (2).¹¹ Although there is inevitably a slight arbitrariness in reducing very general models down to specific simplifications like (3) and (4), the qualitative features of (3) and (4) of most interest here were observable in virtually all the specifications we examined. These are: that changes in patents are noisier and slightly harder to predict than changes in innovations; that the addition of terms in Δy_{t-1} weakens the size of the coefficients on lagged dependent variables and increases the imprecision of their estimates (relative to virtually all AR models); that Δy_{t-1} has a strong but imprecisely estimated effect on changes in innovations/patents; and that the error correction terms identify a strong, relatively precisely estimated effect which works to maintain a stable long-run relationship between the number of innovations and patents and the level of economic activity in the long run.¹²

¹¹ For both regressions, we computed Breusch/Godfrey LM, Durbin's h alternative, ARCH and White heteroscedasticity statistics. In no case did we detect any systematic movement in the residuals. For equation (3) with p values in parentheses; Breusch/Godfrey AR/MA(1) = 0.336 (0.562), Durbin's h alt. = 0.41 (0.682), ARCH = 0.093 (0.761), White = 13.397 (0.495), F (zero slopes) = 10.761 and LLH = -4.131. For (4); Breusch/Godfrey AR/MA(1) = 1.332 (0.248), Durbin's h alt. = -1.043 (0.297), ARCH = 0.000 (0.987), White = 2.69 (0.975), F (zero slopes) = 5.641 and LLH = 18.994. Both regressions are significant improvements over simple AR(1) models (see footnote 7 above) since both ECMs minimise the Akaike Information Criterion relative to their autoregression counterparts: innovations AR(1), AIC = 0.661; innovations ECM (eq (3)), AIC = 0.522; patents AR(1), AIC = -0.619; patents ECM (eq (4)), AIC = -0.857. Results for the truncated sample 1948-79 are qualitatively the same, coefficients in the ECMs are well within their standard errors for the 1948-83 sample and the diagnostics are comparable.

¹² Equation (3) suggests that a 1% permanent increase in demand in period $t-1$ will increase innovations by 1.2% in t and by 2.1% per year thereafter; a 1% permanent increase in demand in period $t-1$ will increase patents by 0.96% in t and 0.31% per year thereafter.

All this seems to add up to the following set of conclusions. Changes in both innovation and patenting activity are pro-cyclical, and both are Granger-caused by changes in output (but do not Granger cause changes in output). Both the number of innovations and the number of patents produced in any year are strongly correlated to the level of economic activity in that year, and, like output, both are non-stationary (at least over the period 1948–83). Both innovations and patents seem to occur in small (several year) clusters which violate the null hypothesis of a random walk. Although these clusters can be accounted for reasonably satisfactorily by using an AR model that does not include any demand variables, the fact is that these clusters are often timed with increases in demand (although the relationship between changes in demand and changes in innovations and patents is hard to estimate precisely).

II.4. *How Robust Are the Results?*

We explored the robustness of these conclusions in four ways. First, as noted above, we applied a fairly wide range of specification tests to a number of different versions of regressions (1)–(4), and almost all of them yielded results qualitatively similar to those reported above.¹³ Second, to test the robustness of our results to the measurement of demand, we repeated the experiments described above using an index of GDP at current factor cost, GDP valued at constant factor cost (in 1985 prices) and an index of non-oil manufacturing output. All of these demand variables produced very much the same results, except that the elasticity of innovations and patents with respect to output in the co-integrating regressions was much larger when the GDP measures were used. Further, changes in output measured in these ways generally had bigger, more precisely estimated effects on changes in innovations and patents in equations like (3) and (4).

Third, our results are robust in the sense of being broadly consistent with previous empirical work. The observation that innovations and patents cluster is not new, although our work has concentrated on clustering at a higher frequency than is usually examined.¹⁴ Further, the suggestion that demand stimulates innovative activity is not new, although the evidence is a little mixed

¹³ In particular, the results are robust to excluding the (possibly contaminated) last four years of data. For the 1948–79 sample our dynamic patents equation is:

$$\Delta p_t = -0.017 - 0.014 \Delta p_{t-1} + 1.412 \Delta y_{t-1} + 1.598 z'_{t-1} \\ (0.475) \quad (0.083) \quad (1.724) \quad (2.255)$$

with diagnostics (p values in parentheses): Breusch/Godfrey AR/MA(1) = 1.332 (0.248); Durbin's h alt. = -0.753 (0.452); ARCH = 1.818 (0.178); White = 5.921 (0.748); F (zero slopes) = 6.173; LLH = 17.611 and $R^2 = 0.407$. Similarly, the dynamic innovations equation is:

$$\Delta n_t = 0.023 - 0.331 \Delta n_{t-1} + 0.975 \Delta y_{t-1} - 0.147 \Delta p_{t-1} - 0.742 z_{t-1} \\ (0.337) \quad (1.782) \quad (0.731) \quad (0.461) \quad (2.569)$$

with diagnostics: Breusch/Godfrey AR/MA(1) = 0.328 (0.567); Durbin's h alt. = -0.044 (0.965); ARCH = 0.006 (0.939); White = 12.86 (0.538); F (zero slopes) = 10.774; $R^2 = 0.624$ and LLH = -2.489.

¹⁴ See Kleinknecht (1981, 1984, 1990), Freeman *et al.* (1982), Solomou (1986) and others for earlier work. In fact, it is possible to interpret the data displayed on Fig. 1 as indicating the existence of a single long-wave cluster of innovations which gradually built up during the late 1950s and early 1960s, and then ended in the later 1970s.

and measured effects are often relatively weak.¹⁵ Finally, the results discussed in this paper are broadly consistent with those reported in the literature on long waves in economic activity (or Kondratiev cycles), namely that innovations tend to cluster during long wave upswings.¹⁶ There is, however, one set of observations which is not consistent with our results. Almost every dating of the Kondratiev cycle identifies a trough in the 1930s, but there are good reasons for thinking that this period saw something of a flowering of inventive activity. Quite why innovative activity flowered in the 1930s is not clear, but it does seem to be the case that this period was something of an exception to the rule that innovative activity is not counter-cyclical.¹⁷

Finally, we were able to examine the robustness of our conclusions to changes in the length of our sample period. The data on major innovations come from a one-off study, and cannot easily be extended forward or backward. The patents data are, however, available from 1883 to 1992, and two interesting observations emerge from examining these data. First, over the very long period, patents appear to be stationary (that is, $I(0)$ rather than $I(1)$) and do not follow a random walk (although they are very highly autocorrelated).¹⁸ They are, however, even more strongly correlated to output over the long term than they appear to be in the shorter post-war sample discussed above (the partial correlation over the period 1883–1992 is 0.9142).¹⁹ Second, the

¹⁵ See Schmookler (1966) for the pioneering work on patents and demand, and Stoneman (1979), Scherer (1982), Walsh (1984), Kleinknecht and Verspagen (1990) and others for cross-section work. Time series work by Saint-Paul (1993) failed to find any cyclical variation in R & D activity (see also Freeman *et al.* (1982)) for a similar conclusion using less formal methods), but Griliches (1990, footnote 20) and Parker (1992) discuss regression results which show some apparently pro-cyclical effects of demand on patenting. Wyatt (1986) reworked some of Schmookler's data and observed causal relations running from investment expenditures to patenting, but not from patents to investment. Mowery and Rosenberg (1979) is a good critical assessment of some of the empirical work on the demand pull hypothesis.

¹⁶ This work dates back to van Gelderen, Kondratiev and Schumpeter. For recent contributions and discussions, see Solomon (1987), Mansfield (1987), Rosenberg and Frischtak (1984), Kleinknecht *et al.* (1992), Berry *et al.* (1993) and the useful summary in Tylecote (1991).

¹⁷ Mensch (1979) has made the strongest case for the view that major depressions trigger innovative activity using somewhat suspect data on 'basic' innovations which occur much less frequently than the SPRU data on innovations or patents (see the careful evaluation of his data in Freeman *et al.* (1982, Chapter 3)). Kleinknecht (1981, 1987 and 1990), Graham and Senge (1980), van Duijn (1981) and others, have discussed further evidence in support of this view, but it hardly adds up to an overpowering case (see the critical remarks and agnostic conclusions of Clark *et al.* (1981) and Solomou (1986)). Tylecote (1991, pp. 65–70) and Freeman *et al.* (1982, pp. 59–63) show that data on 'major' patents over last 200 years are not consistent with Mensch's conclusions. It is possible that more basic or fundamental innovations have different cyclical patterns from less substantive innovations, and this may explain these differences in results.

¹⁸ This unsurprising result is contingent on allowing for the existence of a structural break in the patents and output series for the duration of the Second World War. Otherwise the patents series is trend stationary. Performing a Chow test for the existence of this break in a simple dynamic patenting equation gives a statistic of 25.451 (0.000). A Likelihood Ratio test of the null that the variances over the two sub-samples are different rejects with a statistic of 56.324 (0.000), with p values given in parentheses. Dickey–Fuller tests allowing for this break, as suggested by Perron (1989), reject the existence of a unit root in the patents series. For a ratio of pre-break to total sample (λ) of 0.5, $DF(99, 1) = -4.199$ (10% critical value -3.46) for the change in intercept ('crash') model, $DF(99, 1) = -4.415$ (10% critical value -3.68) for the change in slope ('growth') model, and $DF(99, 1) = -4.169$ (10% critical value -3.96) for the model combining both effects. Critical values are given in Perron (1989), tables IV, V and VI.

¹⁹ Since patents are $I(0)$ and output is $I(1)$ over this longer period, developing an error correction model of changes in output and patents is only of limited interest. However, we repeated most of the tests described above on the longer series, and found that output Granger caused patents (but not the reverse). An analog of (4) which is typical of what we found using regression analysis is: $p_t = 0.431 + 0.884p_{t-1} + 0.058p_{t-2} + 0.24\Delta y_{t-1}$, where $R^2 = 0.912$ and only the co-efficient on p_{t-1} is significant at conventional 5% levels.

relationship between patents and output seems to weaken considerably in the 1980s. An earlier version of this paper worked with patents data for the period 1969–88, and reported a much weaker association between patents and output (the correlation co-efficient is 0.3448), and the complete absence of any causal relationship between the two. This seems to be mainly the result of a sharp plunge in patenting which occurred during the early to middle 1980s while the economy was recovering from the 1980–1 recession. This fall in patenting started in 1979 and mirrors a major fall in the number of innovations produced which also occurred in 1979. It is not hard to believe that both events are a consequence of the exceptionally severe recession which started in 1979, but it is not clear why the recovery in demand in the early to mid 1980s did not stimulate patenting activity in the way it did earlier in the period.²⁰

III. CONCLUSIONS

Some economists regard innovative activity as exogenous, following a ‘supply push’ stochastic trend largely determined by the growth of the stock of knowledge. It is, however, possible to think of a number of reasons why innovative activity might respond to variations in the level of economic activity, and it is certainly not implausible to think that innovative activity might be endogenous. If one adopts the null hypothesis that the intensity of innovative activity varies randomly over time and compares it to the alternative hypothesis that innovative activity fluctuates pro-cyclically, then our results provide reasonably strong (but not overpowering) evidence against the null. Indeed, the data reveal a strong long-run association between the level of innovative activity and the level of demand, a relationship which can be observed at both low and high frequencies (but which may have changed somewhat in the 1980s). Further, changes in demand appear to Granger cause changes in innovative activity, but the data offer no reason for thinking that innovative activity Granger causes changes in demand. We conclude that demand matters, although it is evident that it plays only a relatively modest role in stimulating innovative activity compared to unobserved (and, therefore, apparently stochastic) supply side determinants.

Our data also cast some light on the source of cyclical variations in economic activity, since major innovations and patents are fairly close observable counterparts to the kinds of productivity shocks which real business cycle theorists argue lead to fluctuations in economic activity (e.g Kydland and Prescott (1982), Lucas (1987) and others; Plosser (1989) is a non-technical introduction to real business cycle theory). Although the work that we have reported is in no sense a formal test of real business cycle theory, it is difficult to square our results with the broad spirit of the theory. In particular, it is extremely difficult to discern any causal influence running from observable manifestations of supply side shocks like patents or major innovations to

²⁰ Note that the recession of 1980–1 was exceptionally severe by post-war standards, and it may be that the sharp drop in demand post-1979 disrupted the ‘normal’ structural relationship between the level of innovative activity and demand.

fluctuations in demand in our data. What is more, innovation and patenting activity almost certainly stimulates productivity growth, and the pro-cyclical variations in innovation which we observe are, no doubt, an important contributor to the pro-cyclical variation in productivity growth which has been widely observed. However, the appropriate causal interpretation of this association would seem to run from demand to productivity growth (via innovation) and not the other way round. These conclusions about the sources of cyclical fluctuations in economic activity are, of course, negative, and they do not tell us what does cause booms and slumps. However, the knowledge that variations in the level of innovative activity are probably a consequence and not a cause of fluctuations in demand helps to narrow the search for the true causes.

London Business School

Date of receipt of final typescript: November 1994

REFERENCES

- Arrow, K. (1963). 'Economic welfare and the allocation of resources for innovations.' In *The Rate and Direction of Inventive Activity* (ed. R. Nelson). Princeton: Princeton University Press.
- Banerjee, A., Dolado, J., Galbraith, J. and Hendry, D. (1993). *Co-integration Error Correction and the Econometric Analysis of Non-Stationary Data*. Oxford: Oxford University Press.
- Berry, B., Kim, H. and Kim, H.-M. (1993). 'Are long waves driven by techno-economic transformations?' *Technological Forecasting and Social Change*, vol. 44, pp. 111-35.
- Campbell, J. and Mankiw, N. (1987). 'Are output fluctuations transitory?' *Quarterly Journal of Economics*, vol. 102, pp. 857-80.
- Campbell, J. and Perron, P. (1991). 'Pitfalls and opportunities: what macroeconomists should know about unit roots.' *NBER Macroeconomics Annual*. Cambridge, MA: NBER.
- Clark, J., Freeman, C. and Soete, L. (1981). 'Long waves and technological developments in the 20th century.' *Futures*, August, pp. 308-22.
- Clark, J., Freeman, C. and Soete, L. (1981). 'Long waves and technological developments in the 20th century.' In *Konjunktur, Krise, Gesellschaft* (ed. D. Petzina and G. van Roon). Stuttgart: Klett-Cotta.
- Cochrane, J. (1988). 'How big is the random walk in GNP?' *Journal of Political Economy*, vol. 96, pp. 893-920.
- Dickey, D. and Fuller, W. (1981). 'Likelihood ratio statistics for autoregressive times series with a unit root.' *Econometrica*, vol. 49, pp. 1057-72.
- Dosi, G. (1988). 'Sources, procedures and microeconomic effects of innovation.' *Journal of Economic Literature*, vol. 26, pp. 1120-71.
- Freeman, C., Clarke, J. and Soete, L. (1982). *Unemployment and Technical Innovation*. London: Frances Pinter.
- Gilbert, R. and Newbery, D. (1982). 'Pre-emptive patenting and the persistence of monopoly.' *American Economic Review*, vol. 72, pp. 514-26.
- Graham, A. and Senge, P. (1980). 'A long wave hypothesis of innovation.' *Technological Forecasting and Social Change*, vol. 17, pp. 283-311.
- Griliches, Z. (1990). 'Patent statistics as economic indicators: a survey.' *Journal of Economic Literature*, vol. 28, pp. 1661-707.
- Hall, R. (1991). 'Labour demand, labour supply and employment volatility.' *NBER Macroeconomics Annual*. Cambridge, MA: NBER.
- Jovanovic, B. and Rob, R. (1990). 'Long waves and short waves: growth through intensive and extensive search.' *Econometrica*, vol. 58, pp. 1391-409.
- Judd, K. (1985). 'On the performance of patents.' *Econometrica*, vol. 53, pp. 567-85.
- Kleinknecht, A. (1981). 'Observations on the Schumpeterian swarming of innovations.' *Futures* (August), pp. 293-307.
- Kleinknecht, A. (1984). 'Prosperity, crisis and innovations patterns.' *Cambridge Journal of Economics*, vol. 8, pp. 251-70.
- Kleinknecht, A. (1987). *Innovation Patterns in Crisis and Prosperity*. London: Macmillan.
- Kleinknecht, A. (1990). 'Are there Schumpeterian waves of innovation?' *Cambridge Journal of Economics*, vol. 14, pp. 81-92.
- Kleinknecht, A., Mandel, E. and Wallerstein, I. (eds). (1992). *New Findings in Long-Wave Research*. London: Macmillan.

- Kleinknecht, A. and Verspagen, B. (1990). 'Demand and innovation: Schmookler re-examined.' *Research Policy*, vol. 19, pp. 387-94.
- Kydland, F. and Prescott, E. (1982). 'Time to build and aggregate fluctuations.' *Econometrica*, vol. 50, pp. 1345-70.
- Lee, K., Pesaran, M. and Pierse, R. (1992). 'Persistence of shocks and their sources in a multisectoral model of UK output growth.' *ECONOMIC JOURNAL*, vol. 102, pp. 342-56.
- Lucas, R. (1987). *Models of Business Cycles*. Oxford: Basil Blackwell.
- Mansfield, E. (1983). 'Long waves and technological innovation.' *American Economic Review*, Papers and Proceedings, vol. 73, pp. 141-5.
- Mensch, G. (1979). *Stalemate in Technology: Innovations Overcome the Recession*. New York: Bollinger.
- Mills, T. (1991). 'Are fluctuations in UK output transitory or permanent?' *The Manchester School*, vol. 59, pp. 1-11.
- Mowery, D. and Rosenberg, N. (1979). 'The influence of demand upon innovation: a critical review of some recent empirical studies.' *Research Policy*, vol. 20, pp. 171-8.
- Parker, S. (1992). 'Industrial invention: a supply and demand model for the UK, 1961-1989.' *Applied Economics*, vol. 24, pp. 733-8.
- Penrose, E. (1959). *The Theory of the Growth of the Firm*. Oxford: Basil Blackwell.
- Perron, P. (1989). 'The great crash, the oil price shock and the unit root hypothesis.' *Econometrica*, vol. 57, pp. 1361-401.
- Phillips, P. (1987). 'Time series regression with a unit root.' *Econometrica*, vol. 55, pp. 277-302.
- Phillips, P. and Ouliaris, S. (1990). 'Asymptotic properties of residual-based tests of co-integration.' *Econometrica*, vol. 58, pp. 165-93.
- Plosser, C. (1989). 'Understanding real business cycles.' *Journal of Economic Perspectives*, vol. 3, pp. 51-77.
- Rosenberg, N. (1974). 'Science, invention and economic growth.' *ECONOMIC JOURNAL*, vol. 84, pp. 90-108.
- Rosenberg, N. and Frischtak, C. (1984). 'Technological innovation and long waves.' *Cambridge Journal of Economics*, vol. 8, pp. 7-24.
- Saint-Paul, G. (1993). 'Productivity growth and the structure of the business cycle.' *European Economic Review*, vol. 37, pp. 861-83.
- Scherer, F. M. (1982). 'Demand-pull and technological innovation: Schmookler revisited.' *Journal of Industrial Economics*, vol. 30, pp. 225-37.
- Schleifer, A. (1986). 'Implementation cycles.' *Journal of Political Economy*, vol. 94, pp. 1163-90.
- Schmookler, J. (1966). *Innovation and Economic Growth*. Cambridge, MA: Harvard University Press.
- Solomou, S. (1986). 'Innovation clusters and Kondratieff long waves in economic growth.' *Cambridge Journal of Economics*, vol. 10, pp. 101-12.
- Solomou, S. (1987). *Phases of Economic Growth, 1850-1973: Kondratieff Waves and Kuznets Swings*. Cambridge: Cambridge University Press.
- Stock, J. and Watson, M. (1988). 'Variable trends in economic times series.' *Journal of Economic Perspectives*, vol. 2, pp. 147-74.
- Stoneman, P. (1979). 'Patenting activity: a re-evaluation of the influence of demand pressures.' *Journal of Industrial Economics*, vol. 27, pp. 385-401.
- Townsend, J., Henwood, F., Thomas, G., Pavitt, K. and Wyatt, S. (1981). 'Innovations in Britain since 1945.' Mimeo, Science Policy Research Unit, University of Sussex.
- Tylecote, A. (1991). *The Long Wave in the World Economy*. London: Routledge.
- Van Duijn, (1981). 'Fluctuations in innovations over time.' *Futures*, August, pp. 264-75.
- Walsh, V. (1984). 'Invention and innovation in the chemical industry: demand-pull or discovery-push?' *Research Policy*, vol. 13, pp. 211-34.
- Wyatt, G. (1986). *The Economics of Invention*. Brighton: Wheatsheaf Books.